Imperial College London

BSc/MSci EXAMINATION June 2025

This paper is also taken for the relevant Examination for the Associateship

PRINCIPLES OF INSTRUMENTATION

For 3rd/4th-Year Physics Students

Wednesday, 28$^{\text{th}}$ May 2025: 10:00 to 12:00

The paper consists of **three** *questions – please answer* **all three***.*
A table of Laplace transforms is included at the end of the paper.

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 3 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 3 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. A particle detector contains a liquid scintillator which emits optical photons in response to particle interactions. A fraction of this light is collected by optical fibres immersed in the liquid, with photons detected several metres away by silicon photomultipliers (SiPMs).

(i) (a) List five factors determining the energy threshold of the detector, i.e. the minimum energy deposited in the liquid that allows the particle interaction to be detected.

(b) Explain briefly how a SiPM works, and how its performance compares to that of a photomultiplier tube (PMT) – mention three key parameters.

[10 marks]

(ii) These SiPMs have output resistance of ~300 kΩ, capacitance of 100 pF, and gain $G = 2\times10^6$. Each device is connected to a long transmission line with 50 Ω characteristic impedance and then to an amplifier with 50 Ω input impedance and voltage gain of 20 dB.

(a) Define characteristic impedance Z_0 and state how to calculate this for a transmission line. Do you expect the recorded data to show reflections in the signal? Justify your answer.

(b) At the high frequencies supporting our SiPM signals (~500 MHz) the attenuation of the cable is 0.6 dB/m; what is the reduction in voltage amplitude after 20 m of cable? What causes these losses?

(c) Estimate the voltage amplitude of the single photoelectron response at the amplifier output. What practical source of single photoelectron signals allows us to easily measure this experimentally in SiPMs?

[10 marks]

(iii) A 12-bit analogue-to-digital converter (ADC) is used to measure the amplified SiPM signals with a digital resolution of 1 mV. Determine:

(a) The maximum digitisable input signal in volts.

(b) The dynamic range for the digitised signal in dB.

(c) The maximum quantisation error.

(d) The RMS quantisation noise added.

(e) If the electronics noise from the sensor plus amplifier is 20 μV_{rms} (referred to the amplifier input) what is the signal-to-noise ratio for the measurement of single photoelectron signals?

[10 marks]

(iv) Fast digitisers operating at 500 MS/s record the signal waveforms.

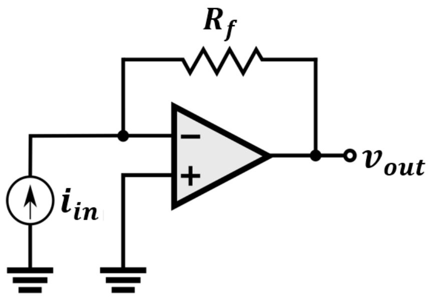
(a) State the Shannon-Nyquist theorem and indicate the Nyquist frequency of the digitised signal.

(b) Discuss briefly what filtering might be required for proper sampling of our signals.

[10 marks]

[Total 40 marks]

2. A transimpedance amplifier (TIA) is often used to convert a low-level current from a photosensor such as a silicon photomultiplier, which generally has high capacitance and high output impedance, into a measurable voltage signal. This is often preferable to converting the current using a simple load resistor R_L. A simplified TIA circuit is shown in the figure below.



(i) Derive the voltage output of the TIA, justifying your assumptions. [5 marks]

(ii) Compare the signal-to-noise ratio for a signal current $i_{in}(t)$ achieved with a TIA and with a load resistor R_L. You may assume that both the current source and the op-amp have negligible intrinsic noise. Explain why using a TIA is nonetheless advantageous, referring to both its input and output impedances. [12 marks]

(iii) Explain why choosing the TIA feedback resistor value is a trade-off between sensitivity and range. How would you determine a reasonable value of R_f knowing that you need to measure currents up to some value i_{max}? Then suggest a trivial modification to the circuit to make it a programmable-gain TIA capable of measuring larger currents. [7 marks]

(iv) From the point of view of bandwidth, what would the advantage be compared with a load resistor given that the photosensor has typically a large capacitance? [6 marks]

[Total 30 marks]

3. (i) A linear sensor is described by the following differential equation, where $y(t)$ is the forcing function and $x(t)$ is the sensor response:

$$a_1 \frac{\mathrm{d}x}{\mathrm{d}t} + a_0 x = b_0 y\,.$$

(a) Show that the transfer function can be written in the form

$$G(s) = \frac{k_s}{\tau s + 1}\,.$$

Express k_s and τ in terms of the constants a_0, a_1 and b_0.

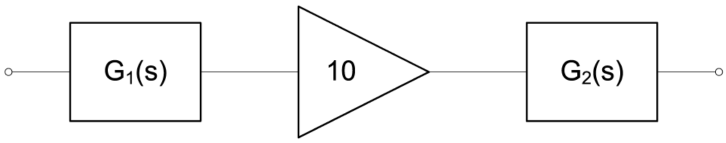
(b) Give the common names for the constants k_s and τ and their physical interpretation in terms of the sensor's time-domain behaviour. How is τ related to the frequency response of the sensor?

(c) For a sensor with $a_0 = 10$, $a_1 = 5$ and $b_0 = 100$ (in appropriate units) determine the sensor's response $x(t)$ to a unit-step at the input. How long does the output take to reach 95% of the steady-state value?

[16 marks]

(ii) In the figure below, a first-order sensor is connected to an ideal amplifier of gain 10 and drives a first-order output transducer. The transfer functions for the first-order elements are:

$$G_1(s) = \frac{10^{-3}}{s + 0.5}, \quad G_2(s) = \frac{20}{s + 2}\,.$$



(a) Write the transfer function for the system. With reference to your answer, does the system behaviour change if the amplifier is moved to the right of the output transducer? Is this physically realistic? Explain your answer.

(b) Determine the system's response to a step function $u(t)$ at the input.

[14 marks]

[Total 30 marks]

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t^n for $n > 0$	$\frac{n!}{s^{(n+1)}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{-at}\sin(\omega t)$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos(\omega t)$	$\frac{s+a}{(s+a)^2+\omega^2}$

Figure 1: Laplace transforms.